



Student Performance Data Analysis and Visualization



Project Overview

Student Performance Data Analysis and Visualization is a Python-based data analysis project that explores student academic and lifestyle-related data using **Pandas** and **Matplotlib**.

The project works with a student dataset containing information such as **study hours, attendance, sleep hours, internet usage, assignments completed, previous scores, exam scores, and placement status**. The data is loaded and analyzed to generate meaningful visualizations that help understand patterns and relationships within student performance.

The main goal of this project is to transform raw student data into understandable visual insights that can help identify factors associated with academic performance and placement outcomes.



Abstract

Student academic performance can be influenced by several factors, including study habits, attendance, sleep patterns, internet usage, assignment completion, and previous academic performance. Analyzing these factors can provide useful insights into student behavior and performance.

This project focuses on analyzing a student dataset using **Python, Pandas, and Matplotlib**. The dataset contains student-related academic and behavioral attributes, including study hours, attendance, sleep hours, internet usage, assignments completed, previous scores, exam scores, and placement status.

The project begins by loading the dataset into a Pandas DataFrame and examining the available student information. Different data visualization techniques are then applied to represent the data graphically. These visualizations make it easier to observe distributions, compare student attributes, and understand relationships between different variables.

The project includes visual analysis of **internet usage and student-related performance data**. By presenting the information through graphs, the project demonstrates how data visualization can be used to convert raw datasets into meaningful and easy-to-understand information.

This project provides practical experience in **data loading, data handling, exploratory analysis, and visualization using Python**. It can also serve as a foundation for developing more advanced student performance prediction or educational analytics systems in the future.

Objectives

- Analyze student academic and behavioral data.
 - Load and process the dataset using Pandas.
 - Explore important student-related attributes.
 - Create meaningful graphs using Matplotlib.
 - Understand patterns in student performance.
 - Present raw data in a simple visual format.
 - Provide a foundation for future predictive analysis.
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Dataset

The project uses a student dataset containing **10,000 student records**.

Dataset Attributes

Attribute	Description
<code>study_hours</code>	Number of hours spent studying
<code>attendance</code>	Student attendance percentage
<code>sleep_hours</code>	Number of hours spent sleeping
<code>internet_usage</code>	Internet usage information
<code>assignments_completed</code>	Number of completed assignments
<code>previous_score</code>	Previous academic score
<code>exam_score</code>	Examination score
<code>placement_status</code>	Student placement status

Technologies Used

- Python
- Pandas
- Matplotlib
- Google Colab
- Jupyter Notebook

The notebook installs and imports Pandas and Matplotlib for data processing and visualization. It also includes the OpenAI Python package in its environment.

Visualizations

The project creates graphical representations of student data using Matplotlib.

Current Visualization

-  Internet Usage Distribution
-  Student-related data visualization
-  Academic performance analysis

The notebook specifically creates a bar chart showing the distribution of `internet_usage` values among students.

Project Workflow

```
graph TD; A[Student Dataset] --> B[Load Dataset]; B --> C[Data Processing using Pandas]; C --> D[Explore Student Attributes]; D --> E[Create Visualizations]; E --> F[Analyze Patterns]; F --> G[Generate Insights];
```

How to Run the Project

1. Clone the Repository

```
git clone https://github.com/your-username/student-performance-analysis.git
```

2. Open the Notebook

Open:

```
Untitled0.ipynb
```

You can run the notebook using:

- Google Colab
- Jupyter Notebook
- JupyterLab

3. Add the Dataset

Place the dataset in the project directory:

```
student_dataset_10000_rows.csv
```

4. Install Required Libraries

```
pip install pandas matplotlib
```

5. Run the Notebook

Execute the notebook cells sequentially to load the dataset and generate the visualizations.

Project Structure

```
Student-Performance-Analysis/  
|  
├── student_dataset_10000_rows.csv  
├── Untitled0.ipynb  
├── README.md  
└── requirements.txt
```

Requirements

Create a `requirements.txt` file containing:

```
pandas  
matplotlib
```

Key Areas of Analysis

The project focuses on understanding:

- Study habits
 - Attendance patterns
 - Sleep duration
 - Internet usage
 - Assignment completion
 - Previous academic scores
 - Examination performance
 - Placement status
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Future Enhancements

The project can be extended by adding:

- Correlation analysis between student attributes.
 - Additional statistical analysis.
 - Interactive dashboards.
 - Student performance prediction.
 - Machine Learning models.
 - Placement prediction.
 - Automated data insights using Generative AI.
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Learning Outcomes

Through this project, the following skills are demonstrated:

- Python programming
 - Data loading and manipulation
 - Pandas DataFrame operations
 - Data visualization
 - Exploratory Data Analysis (EDA)
 - Graph interpretation
 - Working with CSV datasets
 - Google Colab / Jupyter Notebook
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Author

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License

This project is created for **educational and learning purposes**.